STAR-3D: Reliability-Guided RGB-Based Multi-Camera 3D Tracking under Sim-to-Real Shift

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Fig. 1: Multi-camera 3D tracking setting in the AI City Challenge 2026 Track 1 dataset. The central warehouse map shows the positions and viewing directions of four calibrated cameras, with connector lines indicating their corresponding RGB views. Projected ground-truth 3D bounding boxes illustrate how the same warehouse scene is observed under different viewpoints, scales, and occlusion conditions.

Abstract. Multi-camera 3D tracking requires objects observed by several calibrated cameras to be localized and associated within a shared world coordinate system. This task becomes particularly difficult under synthetic-to-real evaluation, where appearance changes and confidence miscalibration can cause missed detections, unstable 3D lifting, and fragmented identities. We present STAR-3D, a reliability-guided RGB-only framework for AI City Challenge 2026 Track 1. STAR-3D combines high-resolution YOLO and D-FINE detectors through class-specific confidence fusion, lifts retained detections using camera calibration and class-level 3D size priors, and consolidates cross-view candidates through reliability-aware multi-view fusion. Class-conditioned online association maintains identities, while conservative BEV tracklet refinement reconnects geometrically and temporally consistent fragments. STAR-3D achieved 12.4128% 3D HOTA, 14.5213% DetA, 11.8109% AssA, and 57.0038% LocA, placing **9th** on the final leaderboard. The results demonstrate the value of propagating reliability across detection, localization, and association under synthetic-to-real shift.

Keywords: Multi-camera 3D tracking · Sim-to-real transfer · Reliability-aware fusion · Bird's-eye-view tracking · Multi-object tracking

1 Introduction

Multi-camera 3D perception is a core requirement for intelligent transportation systems, robotic warehouses, and large indoor smart spaces. In these environments, a single camera rarely observes the full scene, and objects often move across overlapping and non-overlapping fields of view [10,66]. A useful perception system must therefore detect objects in individual images, lift them into a shared world coordinate system, and maintain consistent identities over time [23,24,80]. This requirement makes the task more difficult than standard 2D detection, because success depends not only on recognizing objects, but also on accurate 3D localization and reliable cross-camera association.

AI City Challenge 2026 Track 1 captures this difficulty through a sim-to-real multi-camera 3D tracking setting [2]. The training and validation data are collected from synthetic warehouse environments, while the test data are real-world videos from calibrated multi-camera scenes. Figure 1 illustrates the synthetic multi-camera environment, including the calibrated camera arrangement and the corresponding projected ground-truth 3D boxes. This setting creates a direct domain gap between the visual appearance seen during training and the visual appearance encountered during evaluation [22,67]. At the same time, the output format requires global 3D bounding boxes with object identities, so errors in 2D detection, camera projection, 3D lifting, and temporal association all affect the final 3D HOTA score, which jointly measures detection, association, and localization quality [41]. As a result, the challenge is not only to build a strong detector, but to build a complete perception pipeline that remains reliable when synthetic supervision is transferred to real camera networks.

Existing multi-camera 3D tracking methods often address this problem by combining image detections, camera calibration, depth information, bird's-eye-view aggregation, and tracklet association [32,50,68,72,79,80]. These components are effective when their assumptions are well matched to the data. However, in a sim-to-real setting, each component can introduce a different type of error.

A detector trained on synthetic data may miss rare object classes or produce poorly calibrated confidence scores on real images [20, 32, 47]. A depth-based lifting module may be sensitive to the depth source, camera alignment, or representation mismatch between synthetic and real scenes [50, 68, 72]. A tracker may also create fragmented identities when uncertain detections are discarded or treated in the same way as stable multi-view observations [9,91]. These failure modes suggest that the central problem is not only how to obtain more detections, but how to preserve and use reliability information throughout the full 3D tracking pipeline.

Our work is motivated by this observation. In calibrated multi-camera scenes, object candidates have different levels of geometric support. Some candidates are detected with high confidence, agree across multiple cameras, project consistently into the world frame, and remain stable across nearby frames [23, 24, 80]. Other candidates are visible from only one camera, have low confidence, appear near occlusion boundaries, or produce unstable 3D locations after lifting. Treating these cases equally can increase false positives, fragment identities, and reduce localization quality [9, 79, 91]. A stronger pipeline should therefore propagate reliability from 2D detection to 3D fusion, from 3D fusion to temporal assignment, and from temporal assignment to track birth and track continuation.

Based on this view, we propose STAR-3D, a reliability-guided RGB-only framework for multi-camera 3D tracking under sim-to-real shift. The method first applies high-resolution object detection using complementary detector families, where a recall-oriented detector and a precision-oriented detector are fused to improve coverage while suppressing unstable boxes [49, 63]. The fused 2D detections are then lifted into the global coordinate system using the provided camera calibration and class-specific 3D size priors. During multi-view fusion, each 3D candidate is assigned reliability factors based on detector confidence, camera support, spatial compactness, and consistency with plausible object geometry. These reliability factors are not discarded after fusion. Instead, they are carried into a class-conditioned tracker that adapts association gates, track memory, and birth decisions according to object category and observation stability.

This work makes the following contributions:

1. We formulate sim-to-real multi-camera 3D tracking as a reliability propagation problem, where uncertainty from detection, projection, fusion, and association must be preserved rather than discarded.
2. We propose STAR-3D, a calibrated RGB-only tracking framework that combines high-resolution heterogeneous detector fusion, reliability-guided multi-view 3D lifting, and class-conditioned causal association.
3. We introduce a conservative bird's-eye-view tracklet graph refinement module that reduces identity fragmentation by linking only geometrically and temporally consistent tracklets.
4. We provide an empirical analysis of depth-based and RGB-only strategies for AI City Challenge 2026 Track 1, showing that naive depth injection can

hurt 3D HOTA when depth, calibration, and detector confidence are not jointly calibrated.

2 Related Work

2.1 Calibrated Multi-Camera 3D Perception

Multi-camera 3D perception estimates objects in a shared world frame from synchronized views, and its progress has been driven by autonomous-driving benchmarks [6, 19, 64] and strong LiDAR-based detectors built on voxel, sparse-convolution, pillar, point, center, and transformer representations [31, 45, 60, 61, 86, 88, 94]. Since such methods assume explicit 3D sensing, camera-only approaches instead lift image evidence into 3D, first through monocular depth and box estimation [11, 14, 76–78, 90], and then through multi-camera BEV projection, 3D queries, temporal fusion, and depth-aware supervision [25, 26, 34, 35, 37, 38, 51, 56, 81]. These BEV formulations are further strengthened by cross-modal fusion and distillation, which improve geometric alignment between image features and metric space [13, 36, 39, 40]. Track 1 adds the need for persistent identities, so 3D localization must be coupled with tracking-by-detection methods that use motion, appearance, confidence, and 3D overlap [5, 82, 83, 88, 91, 92], as well as tracklet and BEV-based association methods for longer-term continuity [48, 53, 84]. Recent AI City systems follow this direction with depth maps, point-cloud reconstruction, geometric refinement, view-aware association, and re-identification [33, 50, 68, 73]; in contrast, STAR-3D targets online RGB-only Sim2Real tracking through reliability-aware lifting and conservative BEV association.

2.2 Sim-to-Real Transfer and Reliability Estimation

Sim-to-real transfer is central to Track 1 of AI City 2026 Challenge because models are trained on synthetic warehouse scenes but evaluated on real camera networks. Synthetic and game-engine data provide scalable supervision [16, 54, 57], yet the resulting models often fail when real images differ in texture, lighting, layout, and sensor noise. Domain randomization reduces this gap by broadening the synthetic distribution [52, 58, 67, 69], while adversarial and structured adaptation align synthetic and real data in feature, image, output, or detector space [12, 18, 22, 59, 70, 71]. Self-training and test-time adaptation further use unlabeled target data through pseudo-labels, consistency, entropy minimization, and anti-forgetting updates [46, 62, 74, 75, 85, 87, 95]. However, multi-camera 3D tracking also depends on whether each detection, lifted 3D point, and association is trustworthy. Calibration and uncertainty estimation therefore complement adaptation by identifying miscalibrated, out-of-distribution, and unreliable predictions [17, 20, 21, 27–30, 44, 47]. STAR-3D follows this reliability-centered view by carrying confidence, camera support, spatial compactness, and class priors through RGB-only lifting and association.

2.3 Multi-Object Association and Tracklet Refinement

Multi-object tracking links frame-level detections into identities, so evaluation must capture both localization and association quality [4,41,55,66]. Early tracking-by-detection methods used motion models, appearance embeddings, detector regression, and point-based offsets for online association [3, 5, 83, 93], while later methods learned identity-aware or transformer-based association jointly with detection [7, 43, 65, 89, 92]. Recent online trackers reduce fragmentation by reusing low-confidence detections, correcting motion during occlusion, and adapting re-identification cues [1, 8, 15, 42, 91]. In 3D and multi-camera settings, association can also use metric distance, 3D overlap, velocity, class motion, camera geometry, and tracklet graphs [9, 48, 79, 80, 82, 88]. Recent AI City systems combine these cues with depth, point clouds, geometric refinement, view-aware affinity, and re-identification [33, 50, 68, 73]. STAR-3D follows this direction with conservative online BEV relinking based on motion, class, spatial support, and reliability consistency.

3 Method

3.1 Problem Formulation

Let s denote a multi-camera scene, let $t \in \{1, \ldots, T_s\}$ denote the frame index, and let $c \in \{1, \ldots, C_s\}$ denote the camera index. At time t, the observed RGB frames are $\mathcal{I}_{s,t} = \{I_{s,c,t}\}_{c=1}^{C_s}$. Each camera has known calibration $\Phi_{s,c} = (K_{s,c}, R_{s,c}, o_{s,c})$, where $K_{s,c}$ is the intrinsic matrix, $R_{s,c}$ maps world directions to the camera frame, and $o_{s,c}$ is the camera center in world coordinates. The task is to predict a set of 3D object states in the shared world frame,

$$\hat{\mathcal{Y}}_{s,t} = \{(\hat{k}_i, \hat{a}_i, \hat{q}_i, \hat{d}_i, \hat{\theta}_i)\}_{i=1}^{N_{s,t}}, \tag{1}$$

where $\hat{k}_i \in \mathcal{L}$ is the class label, $\hat{a}_i$ is the predicted track identity, $\hat{q}_i = (\hat{x}_i, \hat{y}_i, \hat{z}_i)$ is the 3D box center, $\hat{d}_i = (\hat{w}_i, \hat{l}_i, \hat{h}_i)$ is the box size, and $\hat{\theta}_i$ is the yaw angle. The label set $\mathcal{L}$ contains the seven challenge classes. In the online setting, $\hat{\mathcal{Y}}_{s,t}$ must be computed from $\{\mathcal{I}_{s,\tau}\}_{\tau \leq t}$, the camera calibration, and parameters estimated from the allowed training and validation data.

3.2 Overview of STAR-3D

STAR-3D follows a tracking-by-detection pipeline, but it keeps reliability information throughout the full system rather than using confidence only as an early filtering score. The method first applies heterogeneous high-resolution detectors to every camera frame. It then lifts each 2D candidate into the common world frame using calibration and class-level size priors. Lifted candidates that agree in bird's-eye view (BEV) are fused into 3D observations, and each observation stores a reliability score derived from detector confidence, number of supporting cameras, cluster compactness, and geometric plausibility. These observations are assigned to active tracks with class-conditioned motion gates. A final conservative BEV tracklet refinement step links only short, geometrically consistent fragments, which reduces identity fragmentation while limiting false merges.

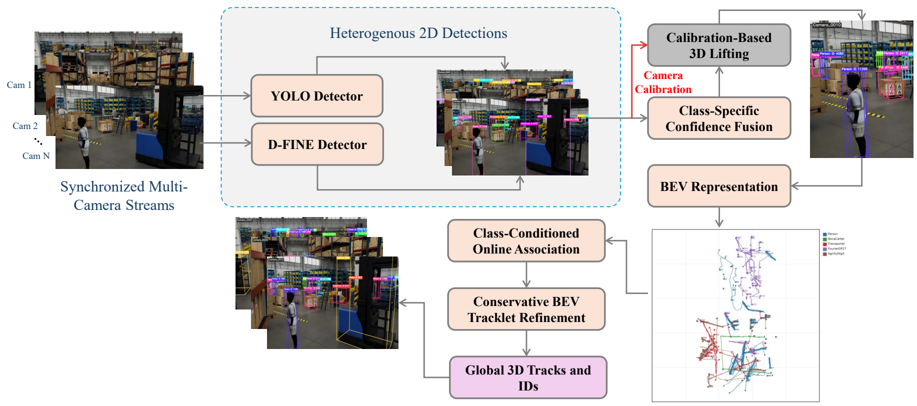

Fig. 2: Overview of the proposed STAR-3D framework for reliability-aware multi-camera 3D tracking.

3.3 High-Resolution Heterogeneous Detection

The detector stage must preserve recall under sim-to-real appearance shift while limiting false positives in crowded warehouse scenes. We therefore use a set of detector models $\mathcal{M}$ rather than a single detector. For model $m \in \mathcal{M}$, camera c, and frame t, the detector output is

$$\mathcal{D}^{m}_{s,c,t} = \{(b_j, k_j, p_j, m)\}_{j=1}^{M^m_{s,c,t}}, \tag{2}$$

where $b_j = (u_{1,j}, v_{1,j}, u_{2,j}, v_{2,j})$ is a 2D bounding box, k_j is the class label, $p_j \in [0,1]$ is the detector confidence, and m records the detector source. In our submitted system, a high-resolution YOLO detector provides broad coverage and a D-FINE detector provides complementary high-precision evidence.

For each camera frame, same-class detections with sufficient 2D overlap are grouped and fused by confidence-weighted box averaging. If $\mathcal{G}$ is one such group, the fused box and confidence are

$$\bar{b} = \frac{\sum_{j\in\mathcal{G}} \beta_{m_j} p_j b_j}{\sum_{j\in\mathcal{G}} \beta_{m_j} p_j}, \qquad \bar{p} = 1 - \prod_{j\in\mathcal{G}} (1 - \beta_{m_j} p_j), \tag{3}$$

where β_{m_j} is a detector-specific calibration weight. The fused detection is written as $\bar{d} = (\bar{b}, \bar{k}, \bar{p}, \bar{m})$, with $\bar{k}$ denoting the shared class of the group and $\bar{m}$ denoting the set of contributing detectors. Class-specific score thresholds are then applied to reduce domination by frequent classes while retaining plausible rare-class candidates.

3.4 Reliability-Aware Multi-View 3D Lifting

Each fused detection is lifted independently before multi-camera fusion. For a detection $\bar{d}$ from camera c, we use the lower center of its image box,

$$\tilde{u}(\bar{d}) = \left(\frac{\bar{u}_1 + \bar{u}_2}{2}, \bar{v}_2, 1 \right)^{\top}, \tag{4}$$

because this point is more stable for grounded objects than the box center. The corresponding world ray is

$$r_{\bar{d}}(\lambda) = o_{s,c} + \lambda R_{s,c}^{\top} K_{s,c}^{-1} \tilde{u}(\bar{d}), \qquad \lambda > 0. \tag{5}$$

Let $\mathcal{P}_s$ be the scene floor plane and let n_s be its upward normal. The ray-plane intersection gives a ground contact point $g_{\bar{d}} = r_{\bar{d}}(\lambda^*)$, where λ^* satisfies $r_{\bar{d}}(\lambda^*) \in \mathcal{P}_s$. We estimate the 3D centroid and box dimensions using robust class priors $\mu_k = (\mu_k^w, \mu_k^l, \mu_k^h)$ computed from the train and validation annotations:

$$q_{\bar{d}} = g_{\bar{d}} + \frac{1}{2} \mu_{\bar{k}}^h n_s, \qquad d_{\bar{d}} = \mu_{\bar{k}}. \tag{6}$$

The candidate yaw $\theta_{\bar{d}}$ is initialized from the recent track direction when the candidate is assigned to a track; otherwise it is set to a class prior.

Lifted candidates are clustered across cameras in BEV space. We write $\psi(q)$ for the projection of a 3D center onto the floor plane. A cluster $\mathcal{H}$ contains same-class candidates whose pairwise BEV distances are below a class-specific gate r_k. For each candidate $\bar{d} \in \mathcal{H}$, its pre-fusion reliability is

$$\rho_{\bar{d}} = \bar{p}^{\alpha} \exp\left(-\frac{\Delta_{\text{size}}(\bar{d})}{\tau_d} \right), \tag{7}$$

where α controls confidence sharpness and τ_d is a size-prior temperature. The term $\Delta_{\text{size}}(\bar{d})$ measures the normalized deviation between the 2D box geometry implied by $\mu_{\bar{k}}$ and the observed image box. The fused 3D observation O for cluster $\mathcal{H}$ is then

$$q_O = \frac{\sum_{\bar{d} \in \mathcal{H}} \rho_{\bar{d}} q_{\bar{d}}}{\sum_{\bar{d} \in \mathcal{H}} \rho_{\bar{d}}}, \qquad d_O = \mu_{k_O}, \tag{8}$$

where k_O is the cluster class. Its cluster spread and camera support are

$$\sigma_O = \frac{1}{|\mathcal{H}|} \sum_{\bar{d} \in \mathcal{H}} \left\| \psi(q_{\bar{d}}) - \psi(q_O) \right\|_2, \qquad \nu_O = \left| \{ c(\bar{d}) : \bar{d} \in \mathcal{H} \} \right|, \tag{9}$$

where $c(\bar{d})$ is the source camera of $\bar{d}$. The final observation reliability is

$$\rho_O = \left(\frac{\sum_{\bar{d} \in \mathcal{H}} \rho_{\bar{d}}}{|\mathcal{H}|} \right) \left(1 - e^{-\nu_O / \tau_c} \right) \exp\left(-\frac{\sigma_O}{\tau_\sigma} \right), \tag{10}$$

where τ_c and τ_σ control the effect of camera support and cluster compactness. Thus each observation is represented as $O = (k_O, q_O, d_O, \theta_O, \rho_O, \nu_O, \sigma_O)$, where θ_O is the confidence-weighted circular mean of the candidate yaw values in $\mathcal{H}$.

3.5 Class-Conditioned Online Association

At frame t, the tracker maintains active tracks $\mathcal{T}_{t-1} = \{T_i\}_{i=1}^{N_{t-1}^{\text{trk}}}$. Each track stores its identity a_i, class posterior π_i, most recent center q_i, velocity v_i, box size d_i, yaw θ_i, reliability history, and inactive age. Before matching, the track center is predicted with a constant-velocity model,

$$\tilde{q}_{i,t} = q_{i,t-1} + v_{i,t-1}\Delta t, \tag{11}$$

where Δt is the frame gap since the last assigned observation. This simple prediction is sufficient for short warehouse-frame intervals and avoids adding a motion model that may overfit the synthetic domain.

For track T_i and observation O_j, we define the association cost

$$\begin{aligned} C_{ij} = \lambda_q \frac{\|\psi(\tilde{q}_{i,t}) - \psi(q_{O_j})\|_2}{r_{k_j}} + \lambda_d \frac{\|d_i - d_{O_j}\|_1}{\|\mu_{k_j}\|_1} \\ + \lambda_\theta \left(1 - \cos(\theta_i - \theta_{O_j})\right) - \lambda_\rho \rho_{O_j} - \lambda_\pi \pi_i(k_j), \end{aligned} \tag{12}$$

where $k_j = k_{O_j}$, and $\lambda_q, \lambda_d, \lambda_\theta, \lambda_\rho, \lambda_\pi$ are non-negative weights. Candidate pairs are invalidated when their classes are incompatible, their BEV distance exceeds the class gate r_{k_j}, or their cost exceeds a class-specific threshold κ_{k_j}. The remaining assignments are solved by Hungarian matching.

After matching, each assigned track is updated with the matched observation and its class posterior is smoothed over time. Unmatched observations start new tracks only when their reliability exceeds a class-specific birth threshold τ_k^{birth}. Unmatched tracks remain active for a class-specific maximum age A_k. These class-conditioned parameters reflect the different behavior of the challenge objects: person tracks use tighter gates and shorter memory, while forklifts, pallet trucks, and mobile robots use wider gates and longer memory because their projected centers are more stable and their occlusions often last longer.

3.6 Conservative BEV Tracklet Refinement

Online association can still fragment identities when a detection is missed for a short interval or when an object changes camera support. We therefore add a conservative BEV tracklet refinement step. A tracklet τ is a maximal sequence of observations assigned to the same identity by the online tracker. We summarize it by

$$\tau = (a_\tau, k_\tau, t_\tau^s, t_\tau^e, q_\tau^s, q_\tau^e, v_\tau, d_\tau, \bar{\rho}_\tau, h_\tau), \tag{13}$$

where a_τ is the track identity, k_τ is the class, t_τ^s and t_τ^e are the start and end frames, q_τ^s and q_τ^e are the first and last 3D centers, v_τ is the average BEV velocity,

d_τ is the average box size, $\bar{\rho}_\tau$ is the average reliability, and h_τ is the normalized camera-support histogram.

Two tracklets τ_a and τ_b are considered only if $k_{\tau_a} = k_{\tau_b}$ and $0 < t^s_{\tau_b} - t^e_{\tau_a} \leq \gamma_k$, where γ_k is a class-specific maximum gap. Let

$$\tilde{q}_{a\to b} = q^e_{\tau_a} + v_{\tau_a}(t^s_{\tau_b} - t^e_{\tau_a}) \tag{14}$$

be the extrapolated endpoint of τ_a at the start time of τ_b. The tracklet-link cost is

$$\begin{aligned} G_{ab} = {} & \eta_q \frac{\|\psi(\tilde{q}_{a\to b}) - \psi(q^s_{\tau_b})\|_2}{r_k} + \eta_v \frac{\|v_{\tau_a} - v_{\tau_b}\|_2}{s_k} \\ & + \eta_d \frac{\|d_{\tau_a} - d_{\tau_b}\|_1}{\|\mu_k\|_1} + \eta_h \left(1 - h^\top_{\tau_a} h_{\tau_b}\right) + \eta_t \frac{t^s_{\tau_b} - t^e_{\tau_a}}{\gamma_k} - \eta_\rho \frac{\bar{\rho}_{\tau_a} + \bar{\rho}_{\tau_b}}{2}, \end{aligned} \tag{15}$$

where $k = k_{\tau_a} = k_{\tau_b}$ and s_k is a class-specific speed normalizer. The coefficients $\eta_q, \eta_v, \eta_d, \eta_h, \eta_t, \eta_\rho$ are non-negative weights. A link is accepted only when G_{ab} is below a strict threshold and no competing link has lower cost for either tracklet. This rule favors short, smooth, reliable connections and rejects ambiguous merges.

The final output is written in the official Track 1 format,

```
scene_id, class_id, object_id, frame_id, x, y, z,
width, length, height, yaw.
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The detector stage mainly affects DetA, the calibrated lifting stage affects LocA, and the online association and BEV refinement stages affect AssA. This separation makes STAR-3D easy to analyze and allows each component to be improved without changing the full pipeline.

4 Experiments

4.1 Dataset and Evaluation Protocol

We evaluate STAR-3D on AI City Challenge 2026 Track 1, which studies multi-camera 3D perception under sim-to-real shift. The released training and validation data contain synthetic warehouse scenes with calibrated multi-camera RGB videos and 3D tracking annotations for seven classes: person, forklift, NovaCarter, transporter, FourierGR1T2, AgilityDigit, and pallet truck. The test split contains real multi-camera videos from related warehouse environments. This split design makes the benchmark challenging because the method must transfer from synthetic visual appearance to real cameras while preserving metric 3D localization and identity consistency.

The official submission requires one text file per scene. Each row contains the scene id, class id, object id, frame id, 3D box center, 3D box size, and yaw angle. We use the online setting for our main submissions, so the prediction at frame t uses only frames up to t. The challenge server reports 3D HOTA as the primary metric, together with DetA, AssA, and LocA, which we define next.

4.2 Evaluation Metrics

The official metric follows the HOTA family, which evaluates detection and association jointly after matching predicted and ground-truth 3D boxes.
3D matching. For a 3D IoU threshold α, let TP_α, FP_α, and FN_α denote true positives, false positives, and false negatives after bipartite matching.
Detection accuracy. Detection accuracy measures whether the submitted boxes recover the correct object set:

$$\mathrm{DetA}_\alpha = \frac{|\mathrm{TP}_\alpha|}{|\mathrm{TP}_\alpha| + |\mathrm{FP}_\alpha| + |\mathrm{FN}_\alpha|}. \tag{16}$$

Association accuracy. For each matched detection $i \in \mathrm{TP}_\alpha$, HOTA also measures whether the matched prediction and ground truth remain associated with the same identities over time. Let $\mathrm{TPA}_\alpha(i)$, $\mathrm{FPA}_\alpha(i)$, and $\mathrm{FNA}_\alpha(i)$ be the true positive, false positive, and false negative association counts for match i. The association term is

$$\mathrm{AssA}_\alpha = \frac{1}{|\mathrm{TP}_\alpha|} \sum_{i \in \mathrm{TP}_\alpha} \frac{|\mathrm{TPA}_\alpha(i)|}{|\mathrm{TPA}_\alpha(i)| + |\mathrm{FPA}_\alpha(i)| + |\mathrm{FNA}_\alpha(i)|}. \tag{17}$$

HOTA. The HOTA score at threshold α is the geometric mean of detection and association accuracy, and the final score averages over the threshold set $\mathcal{A}$ used by the evaluator,

$$\mathrm{HOTA} = \frac{1}{|\mathcal{A}|} \sum_{\alpha \in \mathcal{A}} \sqrt{\mathrm{DetA}_\alpha \mathrm{AssA}_\alpha}. \tag{18}$$

Localization accuracy. The server also reports localization accuracy, which measures the spatial quality of matched 3D boxes. With $\mathrm{IoU}_{3D}(\hat{B}_i, B_i)$ denoting the 3D IoU between a matched prediction and ground-truth box, we write

$$\mathrm{LocA}_\alpha = \frac{1}{|\mathrm{TP}_\alpha|} \sum_{i \in \mathrm{TP}_\alpha} \mathrm{IoU}_{3D}(\hat{B}_i, B_i). \tag{19}$$

These terms are complementary: DetA rewards recovering the correct object set, AssA rewards identity consistency, and LocA rewards accurate 3D box placement. All official results are reported as percentages.

4.3 Implementation Details

Our detector ensemble uses high-resolution YOLO inference at 1920×1920 and a D-FINE precision branch. Detector predictions are fused before 3D lifting, and class-specific score thresholds are tuned on the released validation data. The 3D lifting stage uses the provided calibration and robust class-size priors estimated from the allowed train and validation annotations. Association uses class-specific spatial gates, maximum ages, and cost thresholds. Unless otherwise stated, BEV tracklet refinement uses a maximum gap of 60 frames and a link cost threshold of 1.15. We use the test split only for automatic inference and do not use manual test annotations.

5 Results and Discussion

5.1 Main Results

Table 1 summarizes the official server results for the main submitted systems. The best STAR-3D configuration combines high-resolution YOLO11x detection, D-FINE precision filtering, adaptive online association, and conservative BEV tracklet refinement. This variant obtains the strongest score among our submissions, reaching 12.4128 3D HOTA and ranking ninth on the final leaderboard. The result indicates that our most effective setting is not the one with the highest standalone DetA or LocA, but the one that best balances detection quality with identity continuity.

Table 1: Official Track 1 results for our main submissions. All values are percentages. The best result is shown in bold.

Variant	3D HOTA	DetA	AssA	LocA
YOLO40e + D-FINE + BEV graph	12.0640	13.9304	10.8072	57.0960
YOLO11x-1920 + D-FINE + adaptive	12.3778	14.5214	11.7634	57.0038
YOLO11x-1920 + D-FINE + BEV refinement	**12.4128**	14.5213	**11.8109**	57.0038
YOLO11x-1920 + D-FINE + high-precision	12.1237	14.5781	10.2219	57.0948
YOLO11x-1920 + D-FINE + multi-view optimization	12.1418	**14.5837**	10.2434	**57.1022**
Class-hybrid depth + multi-view optimization	12.1249	14.5835	10.2271	57.1020

5.2 Ablation Analysis

Table 2 isolates the main design choices. Moving from YOLO40e to YOLO11x-1920 improves DetA and raises 3D HOTA, confirming that strong 2D detection is the foundation of the pipeline. Adding conservative BEV refinement gives a smaller but consistent AssA improvement, which shows that short-gap tracklet repair reduces fragmentation without changing the underlying localization. In contrast, the high-precision and depth-hybrid variants improve DetA or LocA but reduce AssA, so their final HOTA is lower. This tradeoff supports the reliability-guided design: a component is useful only when it improves geometry without destabilizing identities.

5.3 Qualitative Results

To qualitatively examine STAR-3D under the real test-domain conditions, Fig. 3 presents the predictions for frame 3000 and the corresponding scene-level tracks. Since the provided test annotations are withheld, the analysis focuses on whether the predicted states remain consistent across camera views, in the shared BEV coordinate system, and over time.

Table 2: Ablation of the main STAR-3D components on the official server.

Variant	3D HOTA	DetA	AssA	LocA
YOLO40e + D-FINE + BEV graph	12.0640	13.9304	10.8072	57.0960
YOLO11x-1920 + D-FINE	12.3778	14.5214	11.7634	57.0038
+ conservative BEV tracklet refinement	**12.4128**	14.5213	**11.8109**	57.0038
+ high-precision detector setting	12.1237	14.5781	10.2219	57.0948
+ depth and class-hybrid correction	12.1249	**14.5835**	10.2271	**57.1020**

The top panel shows the predicted 3D boxes and global identities projected into the synchronized camera views. Despite changes in viewpoint, object scale, and partial visibility, corresponding observations retain the same identity across cameras. This consistency indicates that the calibration-based lifting and reliability-aware multi-view fusion stages link camera-specific detections to a common global object state rather than treating them as independent image observations. The bottom-left panel shows these fused observations in the global BEV at the same frame, where nearby objects remain spatially distinguishable for world-space association. Building on this frame-level representation, the bottom-right panel shows the recovered track histories across the scene. Continuous paths across changing camera coverage illustrate the ability of the class-conditioned association and conservative refinement stages to maintain identity continuity while reconnecting compatible short track fragments. Together, the panels show how STAR-3D transforms multi-view detections into a spatially coherent and temporally consistent global tracking output.

5.4 Overall Challenge Performance

Table 3 presents the official public leaderboard results on the full AI City Challenge 2026 Track 1 test set. STAR-3D ranked **9th** among the listed online submissions, achieving a 3D HOTA of 12.4128%, with DetA, AssA, and LocA scores of 14.5213%, 11.8109%, and 57.0038%, respectively. These results were obtained under the challenging synthetic-to-real evaluation protocol, in which the system was developed using synthetic warehouse data and evaluated on real multi-camera videos.

The metric decomposition shows that STAR-3D localized matched objects more reliably than it detected and associated them. In particular, the LocA score of 57.0038% indicates that the calibration-based RGB lifting produced reasonable 3D locations when detections were correctly matched. In contrast, the lower DetA and AssA scores show that missed detections and track fragmentation remained the main limitations under the synthetic-to-real appearance shift. Since 3D HOTA jointly reflects detection and association quality, these errors limited the overall score despite the comparatively stronger localization performance.

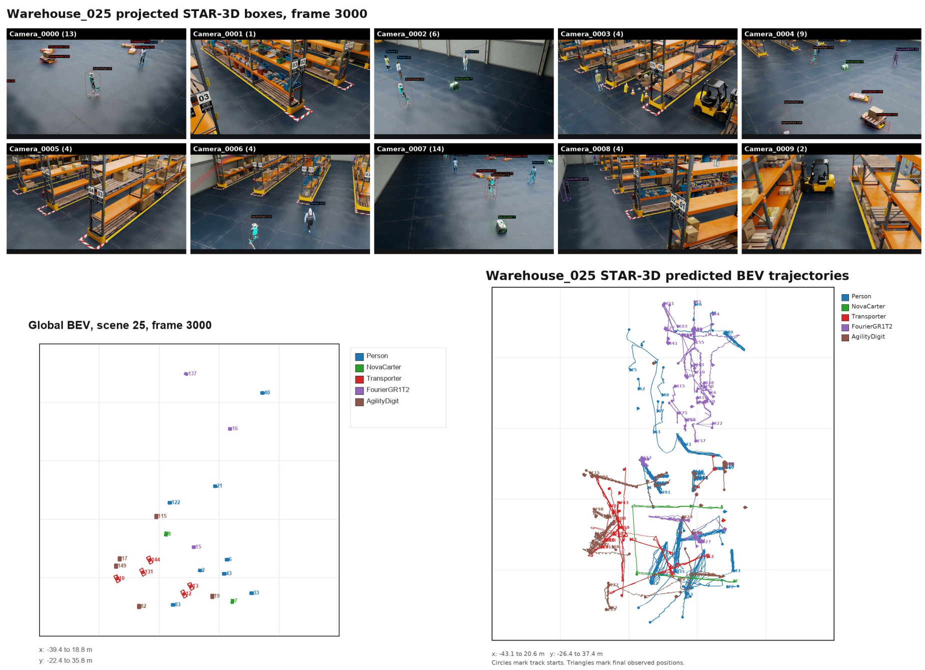

Fig. 3: Qualitative STAR-3D results on a real Track 1 test scene. Top: projected 3D boxes and shared identities across synchronized views. Bottom: the corresponding global BEV state and recovered scene-level track histories. Colors denote object classes.

Table 3: Official public leaderboard results on the full AI City Challenge 2026 Track 1 test set. The STAR-3D submission is shown in bold.

Rank	Method	3D HOTA (%)	DetA (%)	AssA (%)	LocA (%)
1	EVA	56.5447	55.6444	49.3929	79.5558
2	SKKU-AL-T1	52.0118	45.3056	56.5047	76.2410
3	Playbox	38.0105	40.2592	31.0978	75.1778
4	QDTers	34.1845	29.4663	33.1122	17.6841
5	TU-YMLab	25.8837	26.5196	30.6221	69.6980
6	Calix	19.5412	18.6826	18.6161	60.0886
7	mytm	16.6359	16.1613	16.6352	61.6538
8	Anonymous	14.8386	15.1023	15.0593	60.1948
9	**STAR-3D**	**12.4128**	**14.5213**	**11.8109**	**57.0038**

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6 Conclusion

In this work, we presented STAR-3D, a reliability-guided RGB-only framework designed to address the challenges of multi-camera 3D tracking under synthetic-to-real shift. The proposed approach combines complementary high-resolution detectors, class-specific confidence fusion, calibration-based 3D lifting, reliability-aware multi-view fusion, and class-conditioned online association. A conservative BEV tracklet refinement module further reconnects short track fragments only when their geometric, temporal, and reliability characteristics are mutually consistent. Together, these components preserve observation reliability from image-space detection to global 3D localization and identity association.

When evaluated on the AI City Challenge 2026 Track 1 test set, STAR-3D achieved a **3D HOTA score of 12.4128%**, placing **9th on the final leaderboard**. The corresponding DetA, AssA, and LocA scores were 14.5213%, 11.8109%, and 57.0038%, respectively. The comparatively stronger localization performance shows that calibration-based RGB lifting can produce reasonable global 3D positions when objects are successfully detected and matched. At the same time, the lower detection and association scores confirm that missed observations and identity fragmentation remain the primary limitations under domain shift. Overall, the results show that improvements in detection or geometric localization must be jointly designed with temporal association, since gains in one component do not necessarily improve 3D HOTA when identity continuity is weakened.

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